

Facility: Zenith Nexus Core Laboratories

Report Ref. ID: ZN-2026-X77012

Patient ID: P-4451029

Patient Name: Wayne Rooney

Age / Gender: 40 Year(s) / Male

Collected: 15/06/2026 06:00 AM

Received: 15/06/2026 07:45 AM

Reported: 15/06/2026 02:30 PM

PANEL 1: ADVANCED HEMATOLOGY & CELLULAR INDICES (32 Tests)

Complete Blood Count & Differential (CBC-DIFF)

#	Test Parameter	Result	Units	Reference Range	Status
1	Red Blood Cell (RBC) Count	5.12	mill/mm ³	4.50 - 5.50	Normal
2	Hemoglobin (Hb)	17.8	g/dL	13.0 - 17.0	High
3	Hematocrit (HCT / PCV)	47.2	%	40.0 - 50.0	Normal
4	Mean Corpuscular Volume (MCV)	92.1	fL	83.0 - 101.0	Normal
5	Mean Corpuscular Hemoglobin (MCH)	34.8	pg	27.0 - 32.0	High
6	MCH Concentration (MCHC)	33.4	g/dL	31.5 - 34.5	Normal
7	Red Cell Distribution Width (RDW-CV)	12.9	%	11.6 - 14.0	Normal

#	Test Parameter	Result	Units	Reference Range	Status
8	RDW - Standard Deviation (RDW-SD)	41.2	fL	39.0 - 46.0	Normal
9	Total White Blood Cell Count (WBC)	8,450	cells/mm ³	4,000 - 10,000	Normal
10	Mentzer Index	10	Ratio	> 13.0	Low
11	Sehgal Index	1512.4	Ratio	> 972.0	Normal

White Blood Cell Differential (Relative %)

#	Test Parameter	Result	Units	Reference Range	Status
12	Neutrophils Percentage	61.2	%	40.0 - 80.0	Normal
13	Lymphocytes Percentage	26.8	%	20.0 - 40.0	Normal
14	Monocytes Percentage	7.4	%	2.0 - 10.0	Normal
15	Eosinophils Percentage	4.1	%	1.0 - 6.0	Normal
16	Basophils Percentage	0.5	%	0.0 - 2.0	Normal
17	Immature Granulocytes (IG %)	0.2	%	0.0 - 0.5	Normal

White Blood Cell Differential (Absolute Counts)

#	Test Parameter	Result	Units	Reference Range	Status
18	Absolute Neutrophil Count (ANC)	5,171	/mm ³	2,000 - 7,000	Normal
19	Absolute Lymphocyte Count (ALC)	2,264	/mm ³	1,000 - 3,000	Normal

#	Test Parameter	Result	Units	Reference Range	Status
20	Absolute Monocyte Count (AMC)	625	/mm ³	200 - 1000	Normal
21	Absolute Eosinophil Count (AEC)	346	/mm ³	20 - 500	Normal
22	Absolute Basophil Count (ABC)	42	/mm ³	0 - 100	Normal
23	Absolute Immature Granulocytes (AIG)	16	/mm ³	0 - 30	Normal
24	Neutrophil-to-Lymphocyte Ratio (NLR)	2.28	Ratio	1.00 - 3.00	Normal

Platelet & Coagulation Indices

#	Test Parameter	Result	Units	Reference Range	Status
25	Platelet Count	294	10 ³ /μL	150 - 450	Normal
26	Mean Platelet Volume (MPV)	10.6	fL	7.0 - 13.0	Normal
27	Platelet Distribution Width (PDW)	12.4	fL	9.0 - 17.0	Normal
28	Plateletcrit (PCT)	0.311	%	0.200 - 0.500	Normal
29	Large Platelet Cell Ratio (P-LCR)	28.4	%	13.0 - 43.0	Normal
30	Prothrombin Time (PT)	12.1	seconds	11.0 - 14.5	Normal
31	INR (International Normalized Ratio)	1.04	Ratio	0.80 - 1.20	Normal
32	Activated Partial Thromboplastin Time	31.5	seconds	25.0 - 35.0	Normal

PANEL 2: COMPREHENSIVE CARDIO-LIPID METABOLIC MATRIX (22 Tests)

Standard Lipid Profile

#	Test Parameter	Result	Units	Reference Range	Status
33	Total Cholesterol	242	mg/dL	< 200	High
34	Triglycerides	178	mg/dL	< 150	High
35	HDL Cholesterol ("Good")	46	mg/dL	> 40	Normal
36	LDL Cholesterol ("Bad", Calc)	160	mg/dL	< 100	High
37	VLDL Cholesterol (Calculated)	35.6	mg/dL	< 30.0	High
38	Non-HDL Cholesterol	196	mg/dL	< 130	High

Advanced Cardiovascular Biomarkers & Ratios

#	Test Parameter	Result	Units	Reference Range	Status
39	Apolipoprotein A1 (Apo A1)	131.2	mg/dL	105.0 - 175.0	Normal
40	Apolipoprotein B (Apo B)	138.4	mg/dL	60.0 - 140.0	Normal
41	Apo B / Apo A1 Ratio	1.05	Ratio	0.35 - 0.98	High
42	Total Cholesterol / HDL Ratio	5.26	Ratio	3.30 - 4.40	High
43	LDL / HDL Ratio	3.48	Ratio	0.50 - 3.00	High
44	HDL / LDL Ratio	0.29	Ratio	> 0.40	Low

#	Test Parameter	Result	Units	Reference Range	Status
45	Triglyceride / HDL Ratio	3.87	Ratio	< 3.00	High
46	Lipoprotein (a) [Lp(a)]	34.2	mg/dL	< 30.0	High
47	hs-CRP (High Sensitivity CRP)	4.21	mg/L	< 1.00 (Low Risk)	High Risk
48	Homocysteine (Plasma)	14.8	μmol/L	< 15.0	Normal

Glycemic & Pancreatic Markers

#	Test Parameter	Result	Units	Reference Range	Status
49	Fasting Plasma Glucose (FPG)	106	mg/dL	70 - 100	High
50	HbA1c (Glycated Hemoglobin)	6.1	%	< 5.7 (Pre-diab: 5.7-6.4)	High (Pre-Diab)
51	Estimated Average Glucose (eAG)	128	mg/dL	< 117	High
52	Fasting Insulin	14.2	μIU/mL	2.6 - 24.9	Normal
53	HOMA-IR (Insulin Resistance Calc)	3.72	Index	< 2.0	High
54	C-Peptide	2.1	ng/mL	1.1 - 4.4	Normal

PANEL 3: HEPATOBILIARY & EXOCRINE PANCREATIC FUNCTION (20 Tests)

Bilirubin Metabolism

#	Test Parameter	Result	Units	Reference Range	Status
55	Total Bilirubin	0.75	mg/dL	0.20 - 1.30	Normal
56	Direct Bilirubin (Conjugated)	0.18	mg/dL	0.00 - 0.30	Normal
57	Indirect Bilirubin (Unconjugated)	0.57	mg/dL	0.10 - 1.10	Normal

Liver Enzymes

#	Test Parameter	Result	Units	Reference Range	Status
58	AST / SGOT	24	U/L	17 - 49	Normal
59	ALT / SGPT	21	U/L	< 50	Normal
60	AST / ALT Ratio	1.14	Ratio	0.70 - 1.40	Normal
61	Alkaline Phosphatase (ALP)	92	U/L	38 - 126	Normal
62	Gamma-Glutamyl Transferase (GGT)	38	U/L	15 - 73	Normal
63	Lactate Dehydrogenase (LDH)	165	U/L	125 - 243	Normal

Serum Proteins & Enzymes

#	Test Parameter	Result	Units	Reference Range	Status
64	Total Protein	7.8	g/dL	6.0 - 8.3	Normal
65	Albumin	4.6	g/dL	3.5 - 5.0	Normal
66	Globulin	3.2	g/dL	2.3 - 3.5	Normal
67	Albumin / Globulin (A/G) Ratio	1.44	Ratio	0.80 - 2.00	Normal

#	Test Parameter	Result	Units	Reference Range	Status
68	Alpha-1 Antitrypsin	142	mg/dL	90 - 200	Normal
69	Ceruloplasmin	29	mg/dL	20 - 60	Normal
70	Serum Amylase	68	U/L	28 - 100	Normal
71	Serum Lipase	34	U/L	13 - 60	Normal

Bile & Fibrosis Calculations

#	Test Parameter	Result	Units	Reference Range	Status
72	Total Bile Acids	4.5	μmol/L	< 10.0	Normal
73	FIB-4 Index (Fibrosis Calc)	0.82	Score	< 1.45	Normal
74	APRI Score	0.19	Score	< 0.50	Normal

PANEL 4: RENAL ARCHITECTURE, VOLUMETRICS & ELECTROLYTES (18 Tests)

Nitrogenous Waste Products

#	Test Parameter	Result	Units	Reference Range	Status
75	Serum Urea	26.4	mg/dL	19.0 - 43.0	Normal
76	Blood Urea Nitrogen (BUN)	12.3	mg/dL	6.0 - 20.0	Normal
77	Serum Creatinine	0.98	mg/dL	0.66 - 1.25	Normal
78	BUN / Creatinine Ratio	12.55	Ratio	10.00 - 20.00	Normal
79	Serum Uric Acid	6.4	mg/dL	3.5 - 8.5	Normal

Glomerular Filtration & Proteins

#	Test Parameter	Result	Units	Reference Range	Status
80	eGFR (CKD-EPI Formula)	104	mL/min/1.73m ²	> 90	Normal
81	Cystatin C	0.84	mg/L	0.61 - 1.05	Normal
82	eGFR (Cystatin C based)	108	mL/min/1.73m ²	> 90	Normal
83	Beta-2 Microglobulin	1.6	mg/L	0.7 - 3.4	Normal

Mineral & Electrolyte Balance

#	Test Parameter	Result	Units	Reference Range	Status
84	Sodium	141	mmol/L	137 - 145	Normal
85	Potassium	4.4	mmol/L	3.5 - 5.5	Normal
86	Chloride	102	mmol/L	98 - 107	Normal
87	Bicarbonate (Total CO ₂)	25	mmol/L	22 - 29	Normal
88	Anion Gap (Calculated)	14.0	mmol/L	8.0 - 16.0	Normal
89	Serum Total Calcium	9.4	mg/dL	8.4 - 10.2	Normal
90	Ionized Calcium	4.8	mg/dL	4.5 - 5.6	Normal
91	Inorganic Phosphorus	3.6	mg/dL	2.5 - 4.5	Normal
92	Magnesium	2.1	mEq/L	1.5 - 2.5	Normal

PANEL 5: ENDOCRINE SYSTEM & STEROID HORMONE CASCADE (25 Tests)

Pituitary-Thyroid Axis

#	Test Parameter	Result	Units	Reference Range	Status
93	Total Triiodothyronine (T3)	1.12	ng/mL	0.97 - 1.69	Normal
94	Total Thyroxine (T4)	7.80	µg/dL	5.53 - 11.00	Normal
95	Free T3 (FT3)	3.12	pg/mL	2.00 - 4.40	Normal
96	Free T4 (FT4)	1.24	ng/dL	0.93 - 1.70	Normal
97	TSH (Third Generation Assay)	2.640	µIU/mL	0.400 - 4.049	Normal
98	Thyroglobulin	14.2	ng/mL	1.4 - 78.0	Normal
99	Reverse T3 (rT3)	18.4	ng/dL	9.2 - 24.1	Normal

Pituitary-Gonadal & Adrenal Axis

#	Test Parameter	Result	Units	Reference Range	Status
100	Total Testosterone	512	ng/dL	249 - 836	Normal
101	Free Testosterone	11.4	ng/dL	4.8 - 25.7	Normal
102	Bioavailable Testosterone	240	ng/dL	130 - 680	Normal
103	Sex Hormone Binding Globulin (SHBG)	38.5	nmol/L	16.5 - 55.9	Normal
104	Free Androgen Index (FAI)	46.1	%	30.0 - 150.0	Normal
105	Luteinizing Hormone (LH)	4.6	mIU/mL	1.7 - 8.6	Normal
106	Follicle Stimulating Hormone (FSH)	3.8	mIU/mL	1.5 - 12.4	Normal
107	Prolactin	8.4	ng/mL	4.0 - 15.2	Normal

#	Test Parameter	Result	Units	Reference Range	Status
108	Estradiol (E2)	24.5	pg/mL	7.6 - 42.6	Normal
109	Progesterone	0.22	ng/mL	< 0.20 - 0.48	Normal
110	DHEA-Sulfate (DHEA-S)	210.4	µg/dL	106.0 - 464.0	Normal
111	Cortisol (8:00 AM Baseline)	12.4	µg/dL	6.2 - 19.4	Normal
112	Adrenocorticotrophic Hormone (ACTH)	24.1	pg/mL	7.2 - 63.3	Normal
113	Growth Hormone (GH)	0.42	ng/mL	0.05 - 3.00	Normal
114	IGF-1 (Insulin-like Growth Factor)	168	ng/mL	101 - 267	Normal

Bone & Mineral Regulation

#	Test Parameter	Result	Units	Reference Range	Status
115	Intact Parathyroid Hormone (iPTH)	34.2	pg/mL	15.0 - 65.0	Normal
116	Calcitonin	4.1	pg/mL	< 8.4	Normal
117	Osteocalcin	18.5	ng/mL	11.0 - 43.0	Normal

PANEL 6: VITAMIN MICRONUTRIENTS & ANEMIA PROFILES (18 Tests)

Vitamin Biomarkers

#	Test Parameter	Result	Units	Reference Range	Status
118	Vitamin D, 25-Hydroxy	18.4	ng/mL	> 30.0 (Deficient <20)	Low (Deficient)
119	Vitamin B12 (Cyanocobalamin)	388	pg/mL	239 - 931	Normal
120	Folate (Vitamin B9, Serum)	6.4	ng/mL	4.6 - 18.7	Normal
121	Vitamin A (Retinol)	48.2	µg/dL	30.0 - 80.0	Normal
122	Vitamin E (Alpha-Tocopherol)	11.4	mg/L	5.5 - 17.0	Normal
123	Vitamin C (Ascorbic Acid)	0.68	mg/dL	0.40 - 2.00	Normal

Iron Kinetics & Storage

#	Test Parameter	Result	Units	Reference Range	Status
124	Serum Iron	98	µg/dL	65 - 175	Normal
125	Total Iron Binding Capacity (TIBC)	310	µg/dL	250 - 450	Normal
126	Unsaturated Iron Binding (UIBC)	212	µg/dL	131 - 425	Normal
127	Transferrin Saturation Percentage	31.6	%	20.0 - 50.0	Normal
128	Serum Ferritin	114	ng/mL	30 - 400	Normal
129	Serum Transferrin	242	mg/dL	200 - 370	Normal
130	Soluble Transferrin Receptor (sTfR)	1.24	mg/L	1.14 - 1.95	Normal

Trace Heavy Metal Survey

#	Test Parameter	Result	Units	Reference Range	Status
131	Serum Zinc	94.2	µg/dL	70.0 - 120.0	Normal
132	Serum Copper	104.1	µg/dL	80.0 - 155.0	Normal
133	Copper/Zinc Ratio	1.11	Ratio	0.80 - 1.50	Normal
134	Whole Blood Lead	1.4	µg/dL	< 5.0	Normal
135	Blood Mercury	0.8	µg/L	< 5.0	Normal

PANEL 7: IMMUNOGLOBULINS, AUTOIMMUNITY & IMMUNOLOGY (25 Tests)

Humoral Immune Proteins

#	Test Parameter	Result	Units	Reference Range	Status
136	Immunoglobulin G (IgG)	1,140	mg/dL	700 - 1,600	Normal
137	Immunoglobulin A (IgA)	210	mg/dL	70 - 400	Normal
138	Immunoglobulin M (IgM)	94	mg/dL	40 - 230	Normal
139	Immunoglobulin E (IgE, Total)	42	kU/L	< 100	Normal
140	Complement Component C3	118	mg/dL	90 - 180	Normal
141	Complement Component C4	26	mg/dL	10 - 40	Normal

Inflammatory & Autoimmune Serology

#	Test Parameter	Result	Reference Range	Status
142	Rheumatoid Factor (RF)	< 10.0 U/mL	< 14.0	Normal

#	Test Parameter	Result	Reference Range	Status
143	Anti-Nuclear Antibodies (ANA, IFA)	Negative	Negative	Normal
144	ANA Titer (If Positive)	N/A	< 1:40	Normal
145	Anti-dsDNA	8.4 IU/mL	< 25.0	Normal
146	Anti-Cyclic Citrullinated Peptide	4.2 U/mL	< 20.0	Normal
147	Anti-Thyroid Peroxidase (Anti-TPO)	12.4 IU/mL	< 34.0	Normal
148	Anti-Thyroglobulin Antibody (ATG)	14.1 IU/mL	< 115.0	Normal
149	Tissue Transglutaminase IgA (tTG)	2.1 U/mL	< 15.0	Normal
150	Deamidated Gliadin Peptide IgG	3.4 U/mL	< 20.0	Normal

Infectious Disease Serology (Screening Profiles)

#	Test Parameter	Result	Reference Range	Status
151	HBsAg (Hep B Surface Antigen)	Non-Reactive	Non-Reactive	Normal
152	Anti-HCV (Hep C Antibody)	Non-Reactive	Non-Reactive	Normal
153	HIV 1/2 Ag/Ab Combo Screen	Non-Reactive	Non-Reactive	Normal
154	Syphilis RPR Screening	Non-Reactive	Non-Reactive	Normal
155	Helicobacter pylori IgG	Negative	Negative	Normal

#	Test Parameter	Result	Reference Range	Status
156	EBV VCA IgG	Positive	Negative (Indicates past)	Past Immunity
157	EBV VCA IgM	Negative	Negative	Normal
158	CMV IgG	Positive	Negative (Indicates past)	Past Immunity
159	CMV IgM	Negative	Negative	Normal
160	ASO Titer (Antistreptolysin O)	64 IU/mL	< 200	Normal

PANEL 8: EXTENDED ONCOLOGICAL TUMOR MARKERS (10 Tests)

Note: These screening tests are evaluated within context-specific frameworks.

#	Test Parameter	Result	Units	Reference Range	Status
161	PSA, Total (Prostate Specific Ag)	1.24	ng/mL	< 4.00	Normal
162	Free PSA	0.38	ng/mL	Verification Target	Normal
163	Free / Total PSA Ratio	0.31	Ratio	> 0.25	Normal
164	Carcinoembryonic Antigen (CEA)	1.8	ng/mL	< 3.0 (Non-Smoker)	Normal
165	Alpha-Fetoprotein (AFP)	3.2	ng/mL	< 8.4	Normal
166	CA 19-9 (Pancreatic/GI)	12.4	U/mL	< 37.0	Normal

#	Test Parameter	Result	Units	Reference Range	Status
167	CA 125 (Ovarian/General Matrix)	8.4	U/mL	< 35.0	Normal
168	CA 15-3 (Breast/General Matrix)	14.2	U/mL	< 30.0	Normal
169	Beta-HCG (Quantitative Total)	< 1.0	mIU/mL	< 2.0 (Male)	Normal
170	Chromogranin A	42.1	ng/mL	< 93.0	Normal

PANEL 9: PHYSICAL, CHEMICAL & MICROSCOPIC URINALYSIS (30 Tests)

Urinalysis - Physical Qualities

#	Test Parameter	Result	Target Norm	Status
171	Total Urine Volume Submitted	45 mL	Baseline Tracker	Continuous
172	Color Vector Assessment	Pale Yellow	Pale Yellow	Normal
173	Optical Transparency (Clarity)	Clear	Clear	Normal
174	Specific Gravity Index	1.018	1.002 - 1.030	Normal
175	Native pH Quotient	6.2	5.0 - 8.0	Normal

Urinalysis - Automated Strip Chemistry

#	Test Parameter	Result	Target Norm	Status
176	Urinary Protein Matrix	Nil	Nil	Normal
177	Urinary Glucose Matrix	Nil	Nil	Normal

#	Test Parameter	Result	Target Norm	Status
178	Urinary Ketones (Acetoacetate)	Nil	Nil	Normal
179	Urinary Bilirubin Pigment	Nil	Nil	Normal
180	Urinary Urobilinogen Concentration	Normal (0.2)	Normal (< 1.0)	Normal
181	Occult Blood Presence	Negative	Negative	Normal
182	Nitrite Conversion Output	Negative	Negative	Normal
183	Leukocyte Esterase Signatures	Negative	Negative	Normal
184	Urinary Ascorbic Acid	Negative	Negative	Normal

Urinalysis - High-Field Microscopic Morphologies

#	Test Parameter	Result	Units	Reference Range	Status
185	Pus Cells (WBC Count)	1-2	/hpf	0 - 5	Normal
186	Red Blood Cells (RBC Count)	0-1	/hpf	0 - 2	Normal
187	Squamous Epithelial Cells	2-3	/hpf	0 - 5	Normal
188	Renal Tubular Epithelial Cells	Nil	/hpf	Nil	Normal
189	Hyaline Casts	Occasional	/hpf	0 - 2	Normal
190	Granular Casts	Nil	/lpf	Absent	Normal
191	Cellular Casts (WBC/RBC)	Nil	/lpf	Absent	Normal

#	Test Parameter	Result	Units	Reference Range	Status
192	Waxy Casts	Nil	/lpf	Absent	Normal
193	Calcium Oxalate Crystals	Nil	/hpf	Absent	Normal
194	Triple Phosphate Crystals	Nil	/hpf	Absent	Normal
195	Uric Acid Crystals	Nil	/hpf	Absent	Normal
196	Amorphous Urates	Nil	/hpf	Absent	Normal
197	Budding Yeast Cells	Nil	/hpf	Absent	Normal
198	Bacterial Rods / Cocci	Nil	/hpf	Absent	Normal
199	Trichomonas Vaginalis	Nil	/hpf	Absent	Normal
200	Mucus Strands	Absent	/hpf	Absent	Normal

PANEL 10: METABOLIC BASAL FLUID REQUISITIONS (5 Tests)

#	Test Parameter	Result	Units	Reference Range	Status
201	24-Hour Urine Total Protein	84	mg/24h	< 150	Normal
202	24-Hour Urine Creatinine Clear	114	mL/min	90 - 139	Normal
203	Microalbuminuria (Spot)	12.4	mg/L	< 30.0	Normal
204	Urine Creatinine (Spot Reference)	102.4	mg/dL	Verification Index	Normal

#	Test Parameter	Result	Units	Reference Range	Status
205	Albumin-to-Creatinine Ratio (ACR)	12.1	mg/g	< 30.0	Normal